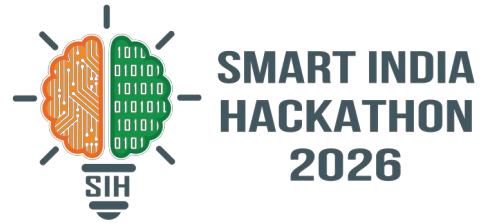


SMART INDIA HACKATHON 2026



KrishiSetu

Problem Statement ID – **SIH26033**

Problem Statement Title – **Farmer-first agricultural trading platform with pooled delivery**

Theme – **Agriculture, FoodTech & Rural Development**

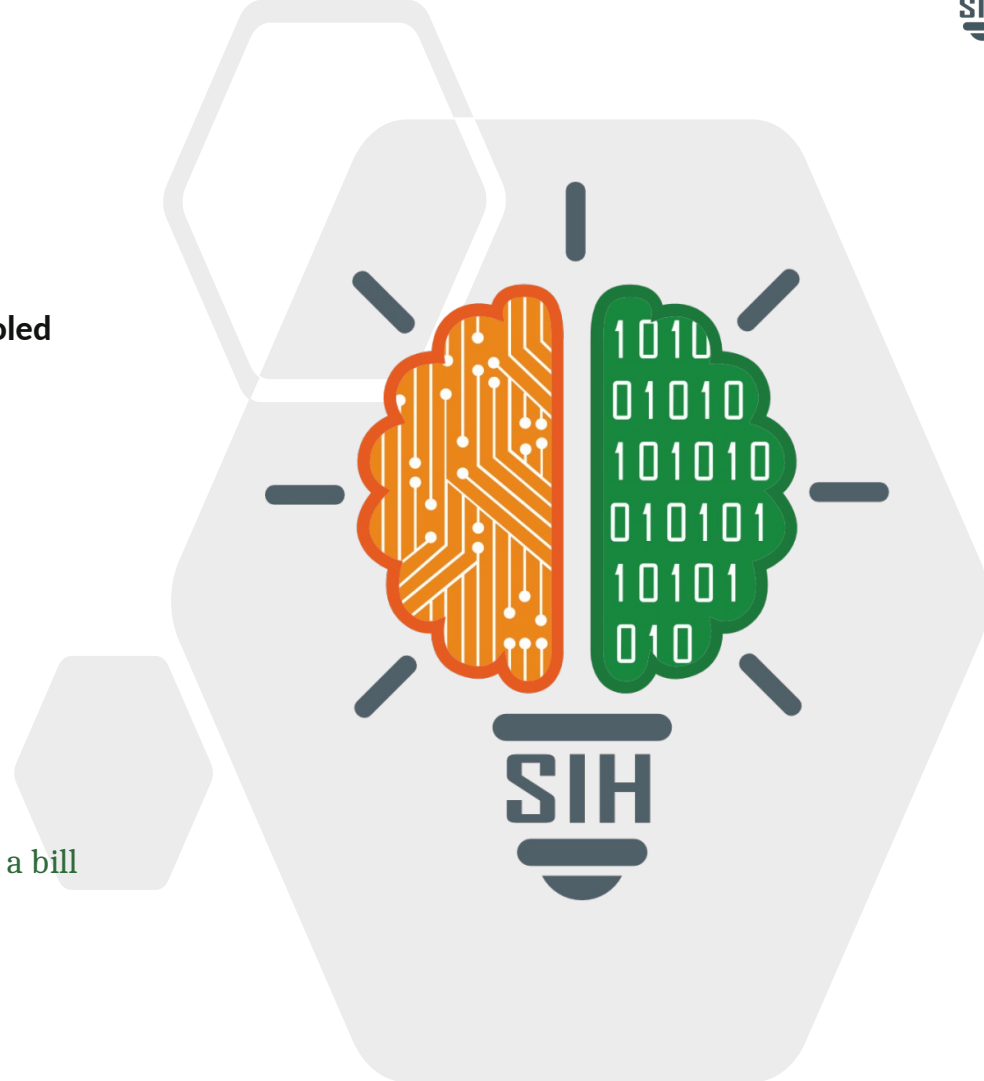
PS Category – **Software**

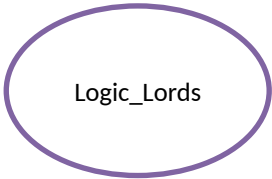
Team ID – **SIH26-SW059**

Team Name – **Logic_Lords — Ramrao Adik Institute of Technology**

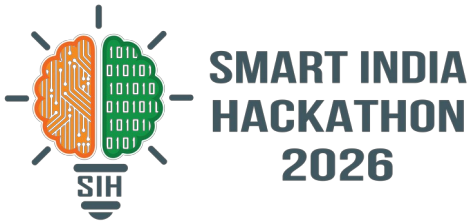
The farmer names their price. Everything after that is shared transport and a bill you can read.

Working prototype live at krishisetu-weld.vercel.app



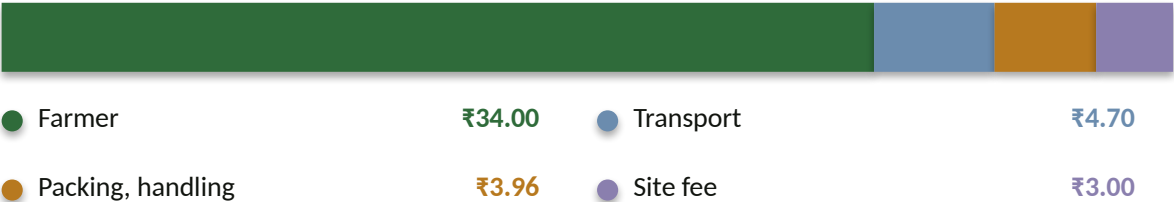


PROPOSED SOLUTION



The farmer's rate is an input to the price, never the leftover

ONE KILOGRAM OF TOMATO, BUILT UPWARD



₹45.66 buyer pays vs ₹55.00 quick-commerce reference

ONE 5 KG BUYER ORDER, FILLED FROM TWO NAMED FARMS

S	Sanjay Patil Khalapur, Raigad	3.0 kg	₹102.00
M	Meena Bhoir Panvel Rural, Raigad	2.0 kg	₹68.00

WHAT IS ACTUALLY DIFFERENT

- 1 Farmer sets the net rate**
No commission, listing charge or logistics cost is deducted from it anywhere.
- 2 Orders are pooled by destination**
Only lots going to the same neighbourhood share a vehicle.
- 3 Nothing moves unsold**
The tempo is loaded against confirmed orders, so there is no stock to write off.

HOW THE BUYER RECEIVES IT

A	Cluster pickup point Kirana or society office holds the batch, 6-9 pm window	₹10
B	Last 300 metres Hand cart inside the same society, free above a ₹500 basket	₹20
C	Express, on demand Not offered — it cannot pay for itself at a 6 kg basket	—

Worked with the same pricing function the live site uses. ₹55/kg is an assumed quick-commerce reference, not a live scrape.

D-COA — Decision & Coordination Optimisation Algorithm

Our working name for the decision layer. It is a product name for our own engine, not a claim of a new published algorithm.



EIGHT CONSIDERATIONS, EVERY TIME A RUN IS PLANNED

● Product compatibility	hard	● Pickup & delivery geography	hard
● Quantity vs vehicle capacity	hard	● Delivery time windows	hard
● Perishability & freshness	hard	● Estimated transport cost	ranking
● Order priority	ranking	● Cancellations & new orders	replan

BUILT AND RUNNING

- Next.js 16 + TypeScript, server components
- PostgreSQL via Prisma, integer paise and grams
- Pooling, routing and forecasting in one codebase
- Live mandi prices from data.gov.in, daily cron
- Deployed on Vercel with Neon Postgres

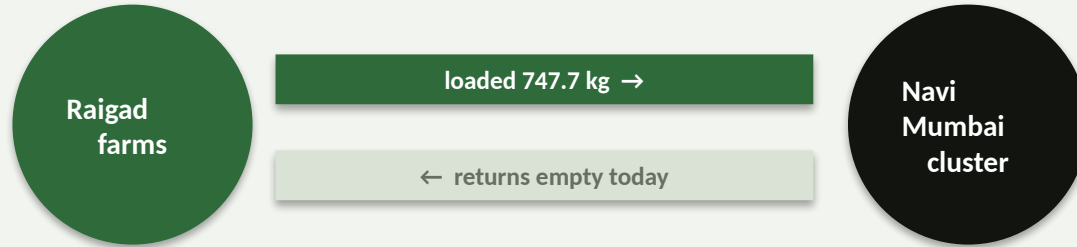
RULES THE CODE ENFORCES

- A run under 70% fill is held, not dispatched
- Reservations are atomic: no kilogram sells twice
- A forecast shows only if it beats a naive baseline

Routing returns a feasible nearest-neighbour plan compared against a fixed-order baseline over the same stops. We do not claim a global optimum.

Return loads, fill thresholds, and what can go wrong

RETURN LOADS — WHY THE EMPTY HALF OF A TRIP MATTERS



A transporter who brings vegetables into the city drives home with an empty vehicle, and that wasted half is already priced into what he charges us. A return load is any cargo he can carry on the way back — seed, fertiliser, packaging, an FPO's supplies — matched to the corridor and time window he has already committed to. He accepts or declines it himself, and the quoted payment must still cover the real detour: empty space does not automatically mean a cheaper trip.

123

orders pooled

747.7 kg

on a 750 kg tempo

5

stops: 4 farms, 1 drop

₹5.37

transport per kg

*One real run, planned by the live system**Return-load matching is designed and specified; it is deliberately after the core proof, and is not claimed as shipped.*

RISK, AND WHAT IS BUILT AGAINST IT



Too few orders in one locality

The run is held below 70% fill and rolls to the next window. A public invite page shows neighbours the exact shortfall.



Produce spoils in transit

Freshness is a hard constraint per commodity; an over-age lot is never loaded on an unrefrigerated run.



Quality disputes

Declared grade, per-stop handover records, and a resolution that can only deduct from a farmer with recorded agreement.

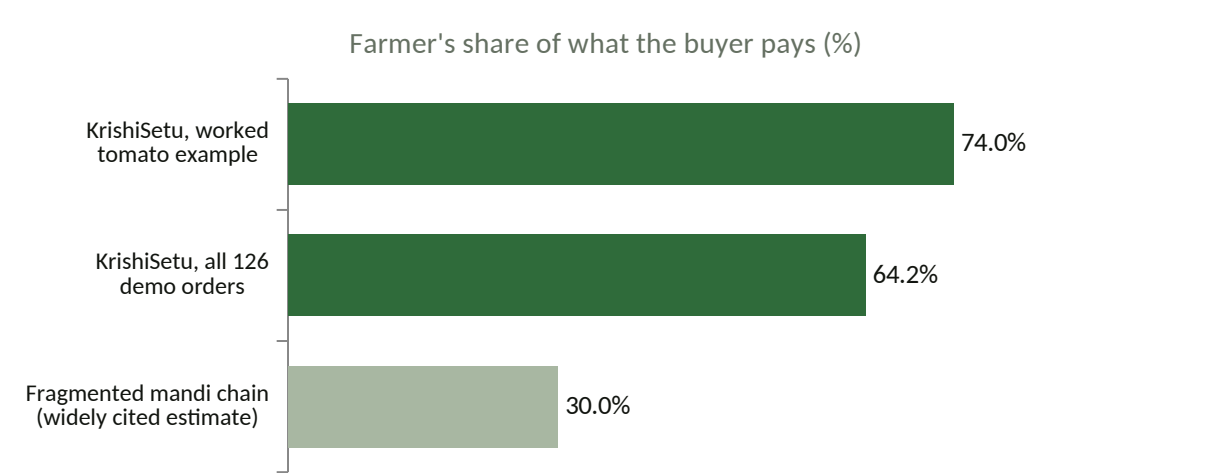


Farmer cannot use an app

Marathi, Hindi and English on the farmer screens; one decision card answers what to send and what it pays.

IMPACT AND BENEFITS

Better for the farmer, cheaper for the buyer, and still self-funding



The fragmented-chain figure is a widely reported range for Indian vegetables, shown as an estimate. The two KrishiSetu bars are computed from orders in the running system.

WHERE THE PLATFORM'S OWN MONEY COMES FROM	
Disclosed buyer-side fee per order	₹19.89
Payment gateway, messaging, support	- ₹5.60
Contribution per completed order	₹14.29
Assumed monthly fixed cost	₹60,000
Orders per month to break even	4,199

No farmer listing fee. No commission on settlement. When a price must be held under the quick-commerce reference, the site fee is cut to zero first — the farmer's accepted amount is never touched.

F

Farmer

₹34.00/kg accepted and paid in full, beside a dated mandi comparison

B

Buyer

₹45.66/kg against a ₹55 reference, every bill line itemised

T

Transporter

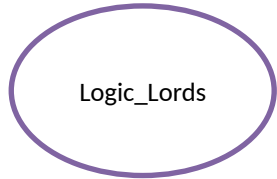
₹4,016 for one full run, agreed before acceptance

P

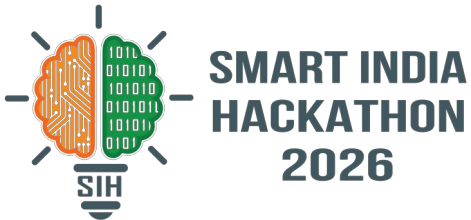
Platform

₹14.29 per order, entirely from a disclosed buyer-side fee

Figures are computed over synthetic demonstration orders, so they size the model rather than measure a business. No production cost is recorded, so no column here is farming profit.



RESEARCH AND REFERENCES



Sources, and what they do and do not prove

Government of India open data portal Daily mandi prices, resource 9ef84268 — the live price feed in the app data.gov.in/catalog/current-daily-price-various-commodities-various-markets-mandi	DATA	eNAM — trading and FPO pages Existing national electronic trade and FPO onboarding enam.gov.in/web/trading-details	INCUMBENT
Ninjacart and Ninja Mandi B2B vegetable supply chain and mandi operations ninjacart.com/ninja-mandi	INCUMBENT	agribazaar Marketplace, logistics and warehousing claims agribazaar.com/quick-links/steps	INCUMBENT
Samunnati FPO financing and market linkage samunnati.com/innovations	INCUMBENT	Google OR-Tools, VRP with time windows Reference formulation for capacitated routing with windows developers.google.com/optimization/routing/vrptw	METHOD
PIB release on agricultural marketing Government position on market reform pib.gov.in — PRID 2222802	POLICY	SIH 2026 idea submission format The template this deck follows sih.gov.in/letters/2026	FORMAT

What this does not prove: demand history and the fitted price series in the prototype are synthetic and labelled so on screen. An unmentioned capability on a competitor's public page is not evidence they lack it, and we claim no measured superiority without a comparable pilot.